OPERATING SYSTEM CONCEPTS - SCHEDULE

Kai Dong, Southeast Univ.

HOUR #	CHAPTER	SECTION	CONCEPT
0∼1	0. Prologue	/	/
7∼10	1. Introduction	What Operating Systems Do Computer-System Organization Operating-System Structure Operating-System Operations	Operating System (system view): resource allocator, control program. Interrupts; Trap/exception. Multiprogramming; Timesharing (multitasking). Interrupt driven; Dual-mode; Privileged instructions.
11∼15	2. Operating-System Structures	System Calls Operating-System Design and Implementation Operating-System Structure	System Calls. Policy and mechanism. Simple / layered / microkernel / modules / hybrid structure.
16∼19	3. Processes	Process Concept Process Scheduling Operations on Processes Interprocess Communication Communication in Client– Server Systems	Processes; Context switch; Process states (transitions); PCB. Long-term / medium / short-term scheduling. Process APIs. Shared memory; Message passing. Pipes.
20∼24	4. Threads	Overview Multicore Programming Multithreading Models Thread Libraries Implicit Threading Threading Issues	Threads; Benefits of threads. Concurrency vs. parallelism. Kernel / user threads; Motivating kernel / user threads; Multithreading models. Pthread APIs. Thread Pools. Thread cancellation; Signal handling; Thread-local storage; Lightweight process.
25∼33	5. CPU Scheduling	Basic Concepts Scheduling Criteria Scheduling algorithms Thread Scheduling Multiple-Processor Scheduling	CPU burst; I/O burst; Scheduling timing; Scheduler; Dispatcher. Scheduling criteria. FCFS; (Preemptive) SJF; Priority; RR; MLQ; MLFQ; Gantt chart; HRRN. Process/system-contention scope. Symmetric multiprocessing; Processor (cache) affinity.
34∼36	6. Process Synchronization	The Critical-Section Problem Peterson's Solution Synchronization Hardware Mutex Locks Semaphores Classic Problems of Synchronization Monitors	Race condition; Critical section; Critical section problem (three requirements). Peterson's Solution. Interrupt masks; Test-and-set; Compare-and-swap; Spin-waiting. Mutex. Semaphores. The Bounded-Buffer / Readers–Writers / Dining-Philosophers problem. Condition variables, Monitors.
37∼42	7. Deadlocks	System Model Deadlock Characterization Methods for Handling Deadlocks Deadlock Prevention Deadlock Avoidance Deadlock Detection Recovery from Deadlock	Deadlock. Four necessary conditions; Resource-allocation graph. Deadlock prevention / avoidance / detection / recovery. Denying the four necessary conditions. Safe state; Resource-allocation graph alg.; The banker's alg. Wait-for graph alg.; Detection alg.. Selecting a victim; Roll back; Starvation.

Continued

HOUR #	CHAPTER	SECTION	CONCEPT
43~50	8. Main Memory	Background Swapping Contiguous Memory Allocation Segmentation Paging Structure of the Page Table	Program loading and linking; Address binding; Logical / physical address space. swapping. base / limit; MMU; Address translation; Free space management. Segmentation; External fragmentation. Paging; TLB; Page size issues; Internal fragmentation. Paging and Segments; Hierarchical page tables; Hashed / inverted page tables.
51~54	9. Virtual Memory	Demand Paging Copy-on-Write Page Replacement Allocation of Frames Thrashing Memory-Mapped Files Other Considerations	Demand paging; Page fault; Effective access time. Copy-on-write. modify / dirty bit; Reference string; Belady's anomaly Page replacement algs. Equal / proportional / priority allocation; Global/local replacement. Thrashing; Working-set model; page-fault frequency. Memory-Mapped Files; Shared memory. Page size issues.
55~57	10. Mass-Storage Structure	Overview of Mass-Storage Structure Disk Structure Disk Scheduling RAID Structure	I/O time. Disk Structure. FCFS; SSTF; SCAN; C-SCAN; LOOK; C-LOOK. RAID lv 0 / 1 / 4 / 5.
58~63	11. File-System Interface	File-System Interface	inode number; File descriptor; Hard / symbolic link; File sharing; Access control.
64	12. File-System Implementation	File Organization Allocation Method Free Space Management	FCB/inode; Direct / indirect pointers; extent. Contiguous / linked / indexed allocation. Bitmaps; Free lists.
64	13. I/O Systems	I/O Systems Application I/O Interface Kernel I/O Subsystem	Polling; interrupt-driven; Direct memory access. Kernel I/O structure. I/O Scheduling; Buffering; Caching; Spooling.